

Research proposal for SME's with High variety and seasonality

Group 6

Target audience

- Mid-sized, seasonal Dutch wholesalers with many product variants — companies facing multi-brand SKU complexity, broken size runs, and seasonal ordering cycles tied to club/league calendars.
- They could increase their market share through technological development.
- A company in the Benelux region.

What they need

- Companies of this nature need a low-cost framework for determining where to deploy AI.
- Key stakeholders need to understand how to leverage AI effectively within their domain.
- Organizations must actively identify and utilize relevant government initiatives.
- To protect limited capital, businesses must establish clear frameworks that distinguish practical AI solutions from market noise.

Research question

- **Main research question:**
 - We wanted to know whether AI-based demand forecasting has actually gotten mature enough for mid-sized, seasonal wholesalers with a lot of product variants to realistically use it alongside the planning tools they already run, or if it's still more promise than something you could actually roll out in the next year or two.
- **Sub-questions:**
 - *Market/external:*
 - *Who is actually building AI demand-forecasting tools for companies this size right now, and are they building for SMEs or mostly chasing the big retail chains?*
 - *Does it matter for this segment whether an AI demand-forecasting vendor is a European supply-chain provider.*
 - *What is the difference for a US or Chinese AI company who operates in the same field?*
 - *Internal:*
 - *What do decision-makers at a company in this segment typically already understand about AI-based forecasting, and is there usually a shared internal sense of what "good enough" historical sales data looks like?*
 - *What do companies that already tried AI-based forecasting say went wrong — and where the available failure-point case studies skew toward*

large retail chains rather than mid-sized seasonal wholesalers with many variants

- *Which of those failure modes (data quality, integration cost, staff distrust of black-box output) are likely structural to AI forecasting generally, versus artifacts of enterprise retail scale?*
- *Where no comparable-scale case studies exist, what does a site visit need to ask directly instead of relying on desk research?*

Quality criteria

- Help readers decide whether to use an existing AI tool, customise current software or develop a new solution. Existing tools should generally be tested before committing to custom development.
- Tailored specifically to the realities of pre-season club cycles, batch manufacturing lead times, and broken size curves rather than generic e-commerce advice.
- Openly details common failure points (e.g., bad historical ERP inputs, black-box distrust from sales staff, rigid vendor contracts) rather than marketing claims.
- Tailored specifically for a business with 100–200 staff and a 1–2 person IT department. Every recommendation must specify transparent price bands (setup cost, monthly SaaS fees, internal hours required) and evaluate off-the-shelf software before suggesting custom development.
- Describe what companies should do after a pilot succeeds, including improving the tool, connecting it to existing systems, training more users and monitoring performance.
- Identify who is responsible for the project, the data, technical implementation, risk management, employee training and final decisions.
- Address privacy, security, bias, transparency, human oversight and accountability. The handbook should make clear which decisions require human review.

Methodology

Desk research

- We are mostly working from what is already out there for this part, vendor sites, product docs, a handful of case studies from companies that tried something similar. That's where we will find out who is actually building AI forecasting tools for companies in our market, versus who is just chasing the big retail chains. It is also the part that tells us what actually goes wrong when people try this in practice, not just what the sales pitch says. And honestly, some of the "how much data do you even need" question can be answered this way too, at least in general terms. There is enough written about typical data requirements that we don't need to figure that out from scratch.

Field research (site visit)

- This is where the real answers are, at least for the two questions that are actually about them specifically. No amount of reading is going to tell us what their sales data actually looks like day to day, or whether they have got someone who could realistically run something like this once it is built, versus needing outside help forever. That's a conversation, not a search.
- Practically, it makes sense to get the desk research mostly out of the way before the visit happens. That way, when we are actually there, we are not wasting time on stuff we could have googled, and we can spend it asking the questions only they can answer.